



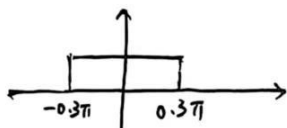
96-22300120273 设计

设计

7.1 画出

$$\text{设 } W_c = \frac{W_p + W_{sc}}{2} = 0.3\pi$$

画出 $H_d(\omega)$



$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & -0.3\pi \leq \omega \leq 0.3\pi \\ 0 & \text{其他} \end{cases}$$

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-0.3\pi}^{0.3\pi} e^{j\omega(n-\alpha)} d\omega \\ &= \frac{1}{2\pi(n-\alpha)} (e^{j0.3\pi(n-\alpha)} - e^{-j0.3\pi(n-\alpha)}) = \frac{2 \sin[0.3\pi(n-\frac{N-1}{2})]}{2\pi(n-\frac{N-1}{2})} \\ &= \frac{\sin[0.3\pi(n-\frac{N-1}{2})]}{\pi(n-\frac{N-1}{2})} \end{aligned}$$

由于阻带衰减要求 -45dB, 选用海明窗

$$\therefore W(n) = [0.54 - 0.46 \cos \frac{2\pi N}{N-1}] \cdot R_N(n)$$

其主瓣宽度为: $\frac{8\pi}{N}$

$$\Delta\omega = (0.4 - 0.2)\pi = 0.2\pi$$

$$\therefore N = \frac{8\pi}{0.2\pi} = 40$$

$$\text{综上, } h(n) = h_d(n) \cdot W(n) = \frac{\sin[0.3\pi(n-\frac{39}{2})]}{\pi(n-\frac{39}{2})} \cdot [0.54 - 0.46 \cos \frac{30\pi}{39}] \cdot R_{40}(n)$$

$0 \leq n \leq 39$

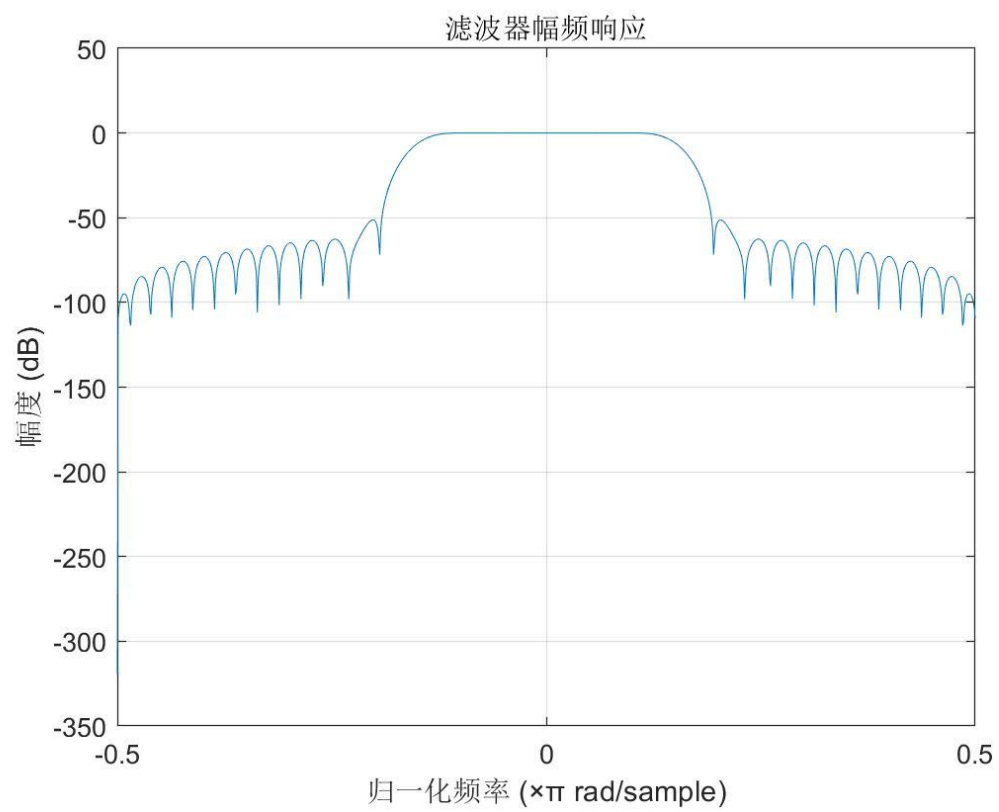


图 1 7.1 的幅频响应

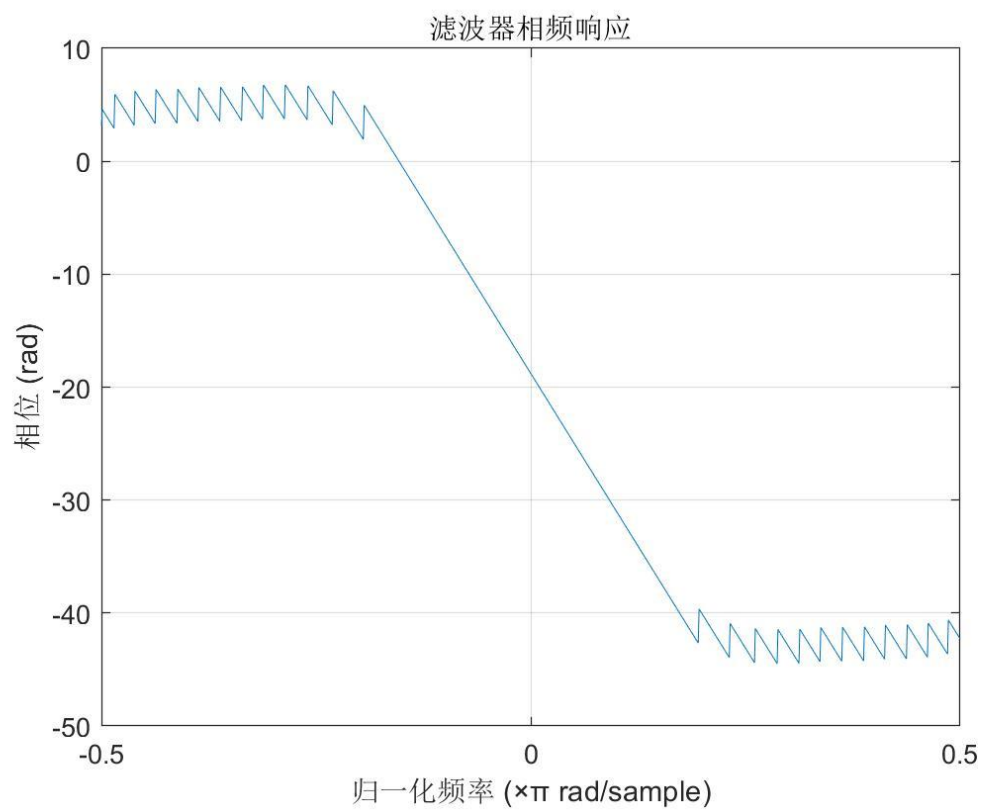


图 2 7.1 相频特性



1.2 解: $\omega_c = (0.1 + 0.5j)\pi = 0.6\pi$

$$H_d(\omega) = \begin{cases} 2e^{-j\omega\alpha} & 0.6\pi \leq |\omega| \leq \pi \\ 0 & \text{其他} \end{cases}$$

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-0.6\pi}^{-\pi} 2e^{-j\omega\alpha} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{0.6\pi}^{\pi} 2e^{-j\omega\alpha} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{-0.6\pi} e^{+j\omega(n-\alpha)} d\omega + \frac{1}{2\pi} \int_{0.6\pi}^{\pi} e^{j\omega(n-\alpha)} d\omega \\ &= \frac{1}{2\pi} \frac{1}{j(n-\alpha)} e^{j\omega(n-\alpha)} \Big|_{-\pi}^{-0.6\pi} + \frac{1}{2\pi} \frac{1}{j(n-\alpha)} e^{j\omega(n-\alpha)} \Big|_{0.6\pi}^{\pi} \\ &= \frac{1}{2\pi} \frac{1}{j(n-\alpha)} (e^{-j\omega 0.6\pi(n-\alpha)} - e^{-j\omega\pi(n-\alpha)}) + \frac{1}{2\pi} \frac{1}{j(n-\alpha)} (e^{j\omega\pi(n-\alpha)} - e^{j\omega 0.6\pi(n-\alpha)}) \\ &= -\frac{\sin[0.6\pi(n-\alpha)]}{\pi(n-\alpha)} + \frac{\sin[\pi(n-\alpha)]}{\pi(n-\alpha)} \end{aligned}$$

$\therefore$ 阻带衰减 $> 55\text{dB}$, 选择布莱克曼窗

$$w(n) = [0.42 - 0.5 \cos \frac{2\pi n}{N-1} + 0.08 \cos \frac{4\pi n}{N-1}] \cdot R_N(n)$$

主瓣宽度为 $\frac{12\pi}{N}$

$$\Delta\omega = 0.2\pi \quad \therefore N = 60$$

要求线性相位, $\alpha = \frac{N-1}{2} = \frac{59}{2}$

综上 $h(n) = h_d(n) \cdot w(n) = \left[\frac{\sin[\pi(n - \frac{59}{2})]}{\pi(n - \frac{59}{2})} - \frac{\sin[0.6\pi(n - \frac{59}{2})]}{\pi(n - \frac{59}{2})} \right] \cdot$

$$\left[0.42 - 0.5 \cos \frac{2\pi n}{59} + 0.08 \cos \frac{4\pi n}{59} \right] \cdot R_{60}(n) \quad (0 \leq n \leq 59)$$

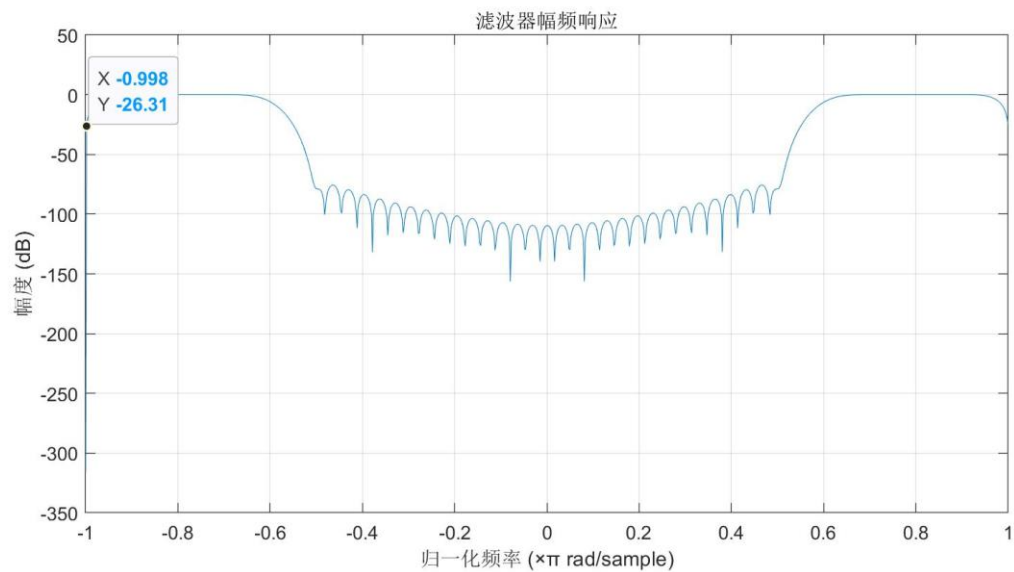


图 3 7.2 幅频响应

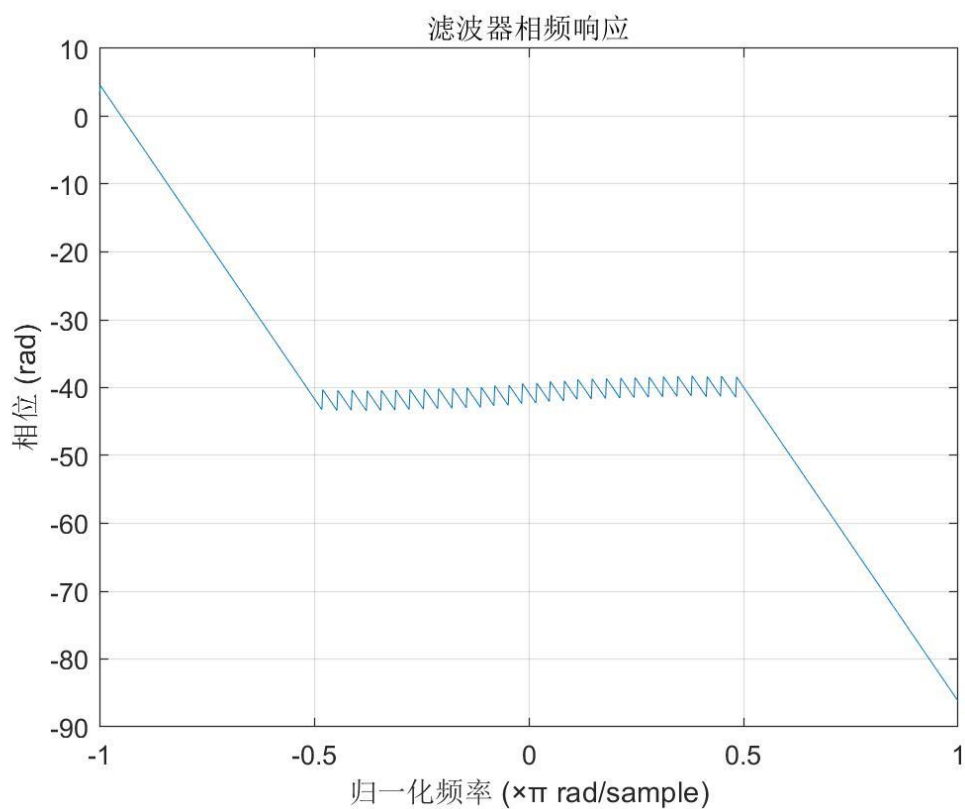


图 4 7.2 相频特性